Roll No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ]

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-4)

MEDICINAL CHEMISTRY-I

Subject Code : BP-402T

M.Code : 75844

Date of Examination : 12-12-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1) Answer briefly :

a) Define bioisosterism.

b) Briefly explain role of hydrogen bonding in drug action with an example.

c) Comment on hydrolytic degradation of acetylcholine.

d) Give typical structural features of a compound to be narcotic analgesic.

e) What are selective COX-2 inhibitors?

f) Give an outline for arachidonic acid pathway by citing the target site of NSAIDs.

g) What are ultra short acting barbiturates?

h) Comment on tissue distribution of adrenergic receptors.

i) Define glucuronidation of drug.

j) What are direct acting sympathomimetic agents?

SECTION-B

2) Explain SAR of sympathomimetic agents. Give mechanism of action and synthesis of Phenylephrine.

3) Explain SAR of benzodiazepine class of sedative hypnotics. Give synthesis of Diazepam.

4) Explain SAR of Morphine analogues. Give synthesis and mechanism of Methadone hydrochloride.

SECTION-C

5) Explain the effect of chelation and partition coefficient on drug action.

6) Discuss the importance of phase II reactions of drug metabolism.

7) Discuss the SAR of beta blockers.

8) Write a short note on synthetic cholinergic blocking agents.

9) Discuss SAR of barbiturates.

10) Describe various classes of NSAIDs with one structural example of each class.

11) Give synthesis and mechanism of Fentanyl hydrochloride.

12) Write a short note on narcotic antagonist.

13) Describe structures and functions of adrenergic neurotransmitters.

Chandigarh College of Pharmacy
LIBRARY
Landran, Mohali

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

EC-2023